National University of Computer & Emerging Sciences - NUCES - Karachi

FAST School of Computing

AI LAB 3 Tasks

Task 1: Goal-Based Navigation Agent

This task involves designing a Goal-Based Agent in a one-dimensional environment where some rooms are dirty. The agent's goal is to clean all rooms and it continues operating until this goal is achieved. At each step, the agent observes the environment, updates its goal if needed, and takes an action. The current percept, goal, and action are displayed at every step.

Task 2: Utility-Based Decision Agent

A Utility-Based Agent selects actions by maximizing utility. The agent can clean the room, move to the next room, or do nothing. Each action is assigned a utility value based on the environment's state. The agent chooses the action with the highest utility and displays the selected action, its utility value, and the total accumulated utility.

Task 3: Learning Agent Performance Improvement

This task focuses on a Learning-Based Agent that improves its behavior over time using rewards. The agent balances exploration and exploitation to learn better actions. As the simulation runs, the agent's performance improves. The final Q-table is displayed to show the learned action values.

Task 4: Multi-Agent Environment Simulation

A Simple Reflex Agent and a Learning-Based Agent operate together in the same environment. Their performance is compared based on task completion speed, action efficiency, and adaptability to environmental changes. This comparison highlights the advantages of learning-based agents in dynamic environments.